Class and Object

Agenda

1. Creating class
2. Creating object
3. Terminologies
4. Constructor
5. Garbage collector
6. Array of objects

1. Creating class

1) Identify the real life entity to model
2) Do its abstraction of state as well as behavior
3) Map attributes of state using variables. Select most suitable data type and give proper names to variables
4) Map behavior by writing functions. Give appropriate name to the function. Decide necessary parameters and return type
5) Bind above variables (state) and functions (behavior) by writing them inside a class. Give appropriate name to the class that indicates the entity it is modeling on our program.
6) Decide what to hide from the code outside the class and make it private

```
class Person {
    private char name[64];
    private int age;
    public char* getName() {
        return name;
    }
    public void setName(char* str) {
        strcpy(name,str);
    }
    public int getAge() {
        return age;
    }
    public void setAge(int a) {
        age=a;
    }
}
```

2. Creating object

OBJECT™ TECHNOLOGIES

- Class v/s object
  - Class is a template that models state and behavior of object. Object is a variable created from the class.
    First class is created. Then objects are created from class
  - Object creation is called as object instantiation
  - Resources are not allocated when class is created. They are allocated to objects that are created from class

- Creating object
  - Person p1; // Object is NOT created. Just a reference is created.
  - p1 = new Person(); // Here object is created and it is referred by reference p1.
  - p1.getName(); // method getName invoked on object p1

3. Terminologies

- Entity
- Class
- Object
- Access specifiers
- Instance and instantiation
- Class members
  - Data members
  - Member functions
  - Accessors
  - Mutators
- Invoking object
- Reference variable and object

4. Constructor

OBJECT™ TECHNOLOGIES

- What is object initialization. Examples of initialization
- Constructor: A member function that initializes object
- Called by compiler, immediately after creation of object
- Rules:
  - Name same as class
  - No return type
- Types:
  - Default constructor
  - Parameter-less constructor
  - Parameterized constructor

4. Constructor

- Constructor overloading
  - Need
  - Same name with different signature
  - Signature is number, type and sequence of parameters.
- Compiler selects definition based on best match of parameters with arguments

5. Garbage collector

OBJECT™ TECHNOLOGIES

- There is no destructor in Java, because we don't delete any object explicitly. We just de-reference it implicitly or explicitly.
- Any un-referenced object is called as garbage
- Garbage collector removes memory for such garbage objects
- Calling garbage collector forcibly
- finalize() method and its role.

6. Array of objects

- Creating array of objects
  Person p[ ]; // p is reference of references
- p= new Person[3]; // 3 references of class Person get created. Person class is not yet instantiated
- p[0]= new Person(); // Here class person is instantiated
- p[1]= new Person("Sachin",52); // Here class person is instantiated
- p[2]= new Person("Rahul",35); // Here class person is instantiated

6. Array of objects

- Iterating an array of objects:
  - Index based iteration

```
for(int i=0; i< pArr.length; i++) {
  pArr[i].display();
}
```

  - Iteration without index

```
for(Person p: pArr) {
   p.display();
}
```

  - Merits and demerits of both types of iteration